

Daejeon-Baro RAG Architecture Validation Report

AI Model & Data Pipeline Feasibility Evaluation Summary

1. Dataset Pipeline Validation

✓ **Verdict: Highly systematic and practically robust design**

- **Clear Separation of Purpose:** Excellently separates the roles of data for LLM tone & manner training (AI-Hub) and fact-based search/reference (Laws & Ordinances).
- **Anti-Hallucination:** Integrating national law databases directly fundamentally blocks the generative AI's biggest weakness—false answers.
- **Localization:** Adding Daejeon's specific ordinances and guidelines alongside central laws perfectly enables practical responses to local complaints (e.g., Tashu, city buses).

2. Stage(1) Intent Classification Validation

✓ **Verdict: Highly standard and valid NLU design**

- **Model Suitability:** Applying the klue/RoBERTa-large model, which excels in context comprehension, is the optimal choice for accurately classifying diverse citizen complaint texts.
- **Practical Exception Handling:** Calculating a Confidence Score alongside the classification result enables sophisticated post-processing logic, such as transferring ambiguous complaints to human agents.
- **RAG Pipeline Efficiency:** Pre-filtering the search scope by the predicted category captures both speed and accuracy, avoiding exhaustive database searches.

3. Stage(2) Dense Retrieval Validation

✓ **Verdict: Perfectly implementable architecture maximizing search accuracy**

- **Optimal Embedding Tool:** Using BGE-m3 (1024-dimensional), which excels in multilingual and long-context processing, accurately maps complex legal texts based on semantics.
- **Standard Preprocessing (Chunking):** Segmenting vast laws into article and clause units for the Vector DB is the textbook method for RAG systems to prevent information dilution.
- **Hybrid Search Synergy:** Utilizing the classification result from Stage 1 as a metadata filter drastically reduces the error of retrieving irrelevant laws.

4. Stage(3) Text Generation Validation

✓ **Verdict: A solid architecture serving as the standard for production RAG systems**

- **Efficient Training (LoRA):** Applying LoRA (Low-Rank Adaptation) fine-tuning to the EEVE model minimizes system resource burden while securing a tone suitable for public institutions.
- **Complete Prompt Engineering:** Perfectly combining [Original Complaint] + [Predicted Category] + [Retrieved Law] flawlessly controls the model's inference direction.
- **Mandatory Citation:** Designing the output to explicitly cite relevant article numbers significantly reduces the time public officials spend verifying facts.



Comprehensive Validation Conclusion

The proposed 'Daejeon-Baro' pipeline is not a simple API-level chatbot. It is a highly cohesive step-by-step architecture flowing from **Classification (NLU)** → **Dense Retrieval** → **Grounded Generation**. Multiple safety mechanisms (metadata filtering, explicit prompting, legal referencing) to control hallucinations are perfectly designed, making it an excellent, practical, and standard system ready for real-world public service deployment.